

# Case 37: Dead-fuel moisture and extinction sweep

## ELMFIRE Verification and Validation Suite

September 2, 2026

### 1 Verification purpose

This case verifies dead-fuel moisture damping, heat of preignition, and extinction for the dead-dominated static fuels FBFM3 and TL6. Moisture is varied as a fraction of each model's configured dead extinction moisture. It can expose percent/fraction errors, incorrect polynomial evaluation, wrong extinction comparison, or omission of moisture from the heat sink. The verification question is whether reaction intensity and ROS follow the independent calculation through extinction for every required variant.

### 2 Mathematical formulation and numerical implementation

For  $r_M = M_f/M_x$ , Rothermel moisture damping is

$$\eta_M = \max\{0, \min[1, 1 - 2.59r_M + 5.11r_M^2 - 3.52r_M^3]\}. \quad (1)$$

Moisture also increases  $Q_{ig} = 250 + 1116M_f$ . Thus it reduces the reaction source and increases the heat sink. The equations follow Andrews (2018), RMRS-GTR-371. ELMFIRE receives percent rasters; the oracle uses fractions.

### 3 Derivation of expected behavior

For each fuel, the 1-h, 10-h, and 100-h moistures are equal, so their weighted dead moisture is the selected ratio times  $M_x$ . The damping polynomial and therefore dead reaction intensity reach zero at and above  $r_M = 1$ . Because these variants have no live load contribution,  $I_R = R_0 = 0$  there. Below extinction, the independent fuel weighting and heat sink give the exact expected curve.

Dead-fuel moisture damping is evaluated from the generated ratio  $r = M_d/M_x$ :

$$\eta_M = \begin{cases} 1 - 2.59r + 5.11r^2 - 3.52r^3, & 0 \leq r \leq 1, \\ 0, & r > 1. \end{cases}$$

For Fuel Model 3 at  $r = 0.50$ ,  $\eta_M = 1 - 2.59(0.5) + 5.11(0.5)^2 - 3.52(0.5)^3 = 0.5425$ . Substitution through the weighted reaction-intensity and Rothermel-rate equations gives  $I_{R,\text{ref}} = 477.3889 \text{ kW m}^{-2}$  and  $R_{\text{ref}} = 1.069455 \text{ m min}^{-1}$ . At  $r = 1.00$ , the polynomial is zero, giving numerical references  $I_R = 3.91 \times 10^{-13}$  and  $R = 6.45 \times 10^{-16}$ , both treated as zero; at  $r = 1.10$  they are exactly zero by the piecewise definition. For nonzero references,  $e_R = |R_{\text{max}} - R_{\text{ref}}| / \max(|R_{\text{ref}}|, 10^{-8})$  and  $e_I$  is analogous. The near-extinction intensity metric instead uses  $\max |I_{R,\text{sim}} - I_{R,\text{ref}}| \leq 0.001 \text{ kW m}^{-2}$ . Propagating variants additionally use  $R_{\text{TOA}} = 60/b$  from  $t = a + bx$  and  $R^2 = 1 - SSE/SST$ , with  $\max e_{\text{TOA}} \leq 0.05$  and  $\min R^2 \geq 0.995$ ; the direct limits are 0.02 for ROS and 0.01 for reaction intensity.

## 4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for direct ROS and nonzero reaction intensity. Analytically extinguished intensity instead uses an absolute tolerance of  $0.001 \text{ kW m}^{-2}$ , so floating-point residuals are not divided by a near-zero reference. Variants predicted to spread at least  $0.10 \text{ m min}^{-1}$  additionally require TOA-derived ROS error at most 0.05 and regression  $R^2 \geq 0.995$ ; slower variants cannot cross three 10 m cells within the capped practical duration and are decided by direct ROS and intensity. Extinguished variants require zero ROS and reaction intensity within the stated absolute residual tolerance; TOA is not used to invent a velocity where no propagation occurs. All 16 variants are mandatory.

## 5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE’s supported  $[-1, 1]$  interval, preserving both an ignited interior and a resolved front gradient. For every positive oracle ROS, `SIMULATION_DT` and `SIMULATION_DTMAX` use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured CFL limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE’s stalled-front branch and corrupt the terminal timestamp. A zero-ROS variant retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. After subtracting the configured meteorology interval, an ordinary stalled-final record must reconstruct the stop time within one timestep. An early stalled-final record is accepted only for a variant whose independent oracle predicts zero ROS, and only when the reconstructed pre-jump time is finite, lies between zero and the stop time, and is aligned to the configured timestep grid. Making TOA optional does not authorize an early stall. The sweep ratios are 0.10, 0.25, 0.50, 0.70, 0.85, 0.95, 1.00, and 1.10. Domains use 10 m cells and a finite ignition strip with a planar east face in the interior measurement corridor; the strip terminates inside the lateral boundaries. Wind and slope are zero, live moisture is fixed, and no coupled fire processes are enabled. Intensity and ROS should decrease toward zero and remain zero beyond extinction.

| Parameter           | Configured value                                          | Units              | Role                            |
|---------------------|-----------------------------------------------------------|--------------------|---------------------------------|
| Domain/grid         | 1400 m square; 10                                         | m/cell             | Planar interior propagation     |
| Fuel models         | FBFM3 and TL6                                             | code/name          | Two extinction-moisture states  |
| Dead-moisture ratio | 0.10 through 1.10 times<br>dead extinction mois-<br>ture  | fraction           | Independent damping variable    |
| Wind/slope          | 0 / 0                                                     | mph / de-<br>grees | Isolates moisture damping       |
| Timing              | generated whole-second<br>step and exact-multiple<br>stop | s                  | Terminal-evidence compatibility |

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — variants/fuel\_186\_r\_010; WS=0 mph, WD=270 deg

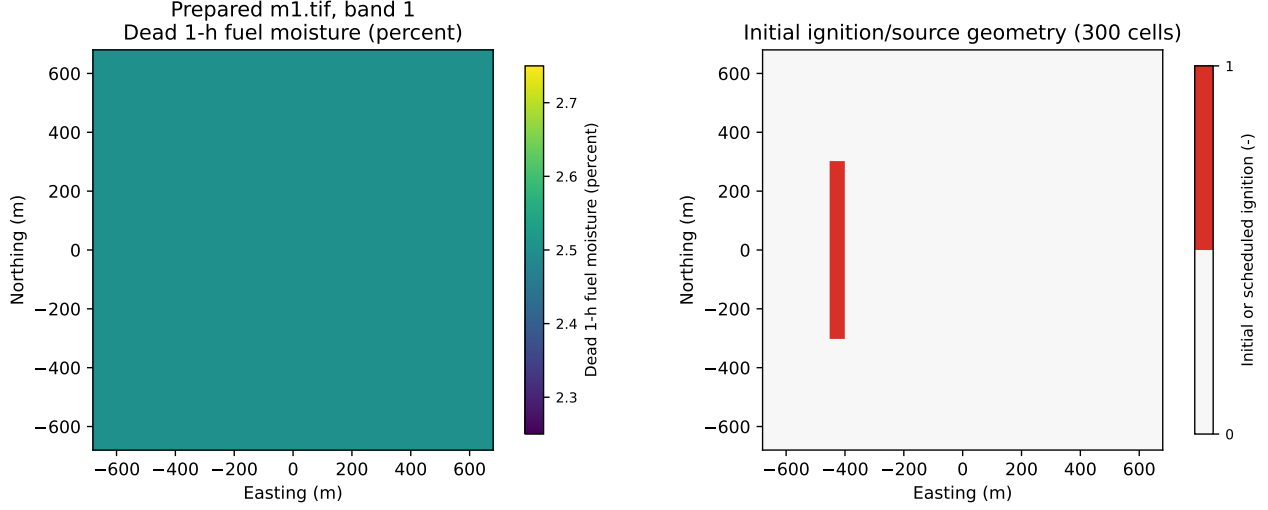


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from  $\phi \leq 0$ .

## 6 Actual simulation results

The generated record contains 16 of 16 required variants and reports **PASS**. For a stalled extinction variant, the machine-readable result records the configured timestep, stop time, meteorology interval, recorded final time, reconstructed pre-jump time, and evidence-selection mode.

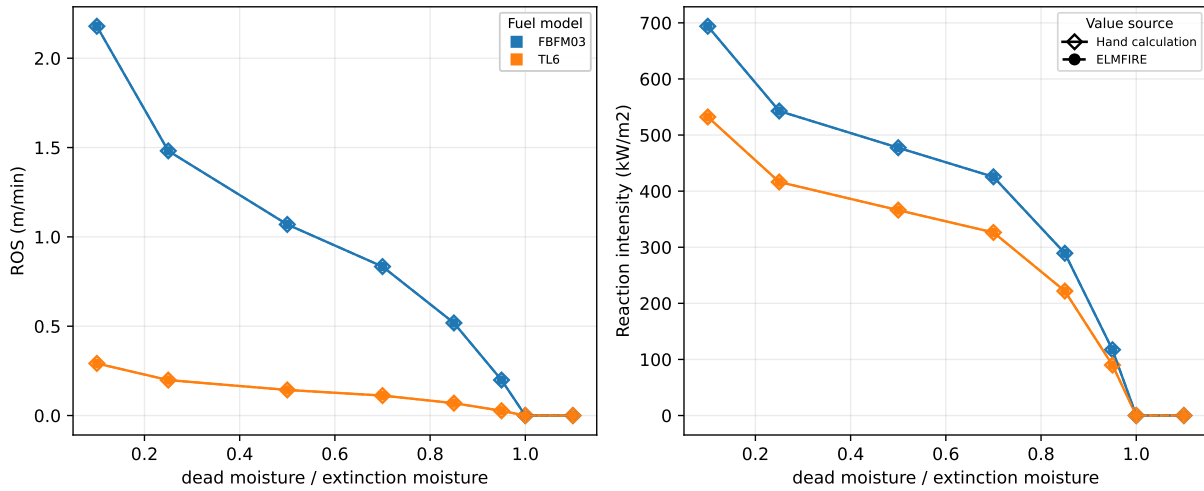


Figure 2: Hand calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines) for the dead-moisture response through extinction. Each fuel-model pair uses the same color.

## 7 Calculated metrics and verification decision

| Metric                                                    | Limit        | Calculated                    | Status |
|-----------------------------------------------------------|--------------|-------------------------------|--------|
| maximum direct-ROS relative error                         | $\leq 0.02$  | 3.66859e-06 fraction          | PASS   |
| maximum reaction-intensity relative error                 | $\leq 0.01$  | 3.31587e-06 fraction          | PASS   |
| maximum near-extinction reaction-intensity absolute error | $\leq 0.001$ | 0.000263903 kW/m <sup>2</sup> | PASS   |
| maximum TOA-ROS relative error                            | $\leq 0.05$  | 0.028024 fraction             | PASS   |
| minimum TOA regression R <sup>2</sup>                     | $\geq 0.995$ | 0.999896 dimensionless        | PASS   |

The authoritative decision is **PASS**.

## 8 Technical interpretation

The PASS decision is driven by the component statuses in the generated table. Direct observables use the upper envelope of the sparse diagnostic raster, excluding zero-filled and off-axis cells from the head-spread comparison. For direct ROS, the largest relative discrepancy is 3.66859e-06. Its independent value is 0.0266137 m/min; ELMFIRE gives 0.0266138 m/min. For reaction intensity, the largest relative discrepancy is 3.31587e-06. Its independent value is 89.9786 kW/m<sup>2</sup>; ELMFIRE gives 89.9789 kW/m<sup>2</sup>. A horizontal shift of the extinction point indicates an  $M_x$  or units error. Correct intensity with wrong ROS suggests heat-sink or flux-ratio handling. Nonzero propagation or intensity at extinction indicates an unintended live contribution, adjustment, numerical residual, or stale output.

## 9 Scope and limitations

The case varies all three dead moisture classes together. It does not identify an isolated class-indexing defect or validate moisture observations. Static live moisture is addressed separately by CASE38.